

ACADIA VALLEY, AB, Canada

WMO: 710480

Lat: 51.066N

Lon: 110.317W

Elev: 2411

StdP: 13.46

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.8	-16.1	-27.0	1.3	-20.5	-21.7	1.8	-15.9	31.2	26.9	27.4	31.3	6.6	270	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	88.2	61.8	84.6	61.2	81.1	60.5	66.0	78.8	64.3	77.2	62.8	76.1	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	90.5	70.8	59.8	83.9	69.3	57.9	78.4	67.1	32.1	78.5	30.7	77.1	29.5	76.6	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.6	24.6	21.2	DB	-28.9	94.2	3.1	3.2	-31.1	96.5	-33.0	98.4	-34.7	100.2	-37.0	102.6	(4)
(5)			WB	-29.1	70.5	3.1	2.2	-31.3	72.1	-33.1	73.3	-34.9	74.6	-37.1	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	14.8	12.1	26.1	40.4	52.6	60.2	65.9	64.5	55.6	41.3	25.6	13.2		
	DBStd	22.45	14.72	14.65	14.66	9.20	8.10	5.93	5.10	5.80	8.88	9.55	13.47	14.28		
	HDD50	5490	1092	1061	741	304	67	5	0	1	45	297	735	1142		
	HDD65	9491	1557	1481	1205	737	389	164	50	82	300	736	1183	1607		
	CDD50	1662	0	0	0	17	147	312	494	451	213	26	2	0		
	CDD65	191	0	0	0	0	5	21	79	67	19	0	0	0		
	CDH74	3049	0	0	0	8	196	355	1090	989	402	9	0	0		
	CDH80	948	0	0	0	0	37	88	354	328	141	0	0	0		
(14) Wind	WSAvg	9.9	10.6	9.7	10.2	11.1	10.5	10.1	8.8	8.4	9.2	10.4	9.8	9.6		
(15) Precipitation	PrecAvg	13.0	0.6	0.3	0.5	0.8	1.5	2.9	2.1	1.7	1.0	0.6	0.5	0.4		
	PrecMax	18.5	1.9	0.7	1.4	2.4	3.1	5.9	5.0	3.3	3.3	1.3	1.7	0.7		
	PrecMin	6.8	0.1	0.0	0.1	0.1	0.1	0.7	0.8	0.2	0.0	0.2	0.0	0.1		
	PrecStd	3.0	0.4	0.2	0.3	0.6	0.9	1.3	0.9	0.8	0.8	0.3	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	45.8	62.5	73.6	84.7	87.8	92.9	91.5	90.1	74.8	63.7	46.8		
		MCWB	39.4	38.6	47.4	50.9	57.6	61.5	65.2	61.5	60.1	54.7	48.0	39.3		
	2%	DB	41.8	40.0	56.2	68.2	79.4	81.9	87.8	87.5	84.7	67.3	53.5	41.3		
		MCWB	36.7	34.9	43.7	48.7	55.6	60.4	63.5	61.1	58.7	51.0	42.6	35.8		
	5%	DB	38.2	36.4	50.3	62.5	75.0	77.7	83.7	83.8	78.5	61.8	47.2	36.7		
		MCWB	34.4	32.9	40.7	45.6	53.6	59.8	63.2	60.8	57.0	48.5	39.0	32.4		
	10%	DB	35.2	33.0	44.4	57.3	69.8	74.3	80.2	79.8	72.9	56.5	41.9	32.3		
		MCWB	32.2	30.5	38.1	43.1	51.4	57.4	62.2	60.2	54.7	45.5	36.5	29.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	39.6	48.0	51.8	58.9	66.4	70.0	66.8	62.1	56.2	49.0	40.3		
		MCDB	46.4	45.3	60.9	71.5	79.0	78.5	85.7	80.2	83.7	71.6	63.1	45.7		
	2%	WB	36.9	35.4	44.5	48.8	56.6	63.4	67.0	64.6	59.7	52.4	43.8	36.0		
		MCDB	41.4	39.1	54.5	66.1	75.4	76.3	79.8	78.4	80.4	64.1	52.6	41.2		
	5%	WB	34.7	33.3	41.3	46.5	54.8	61.1	65.3	63.2	57.9	49.1	39.8	32.9		
		MCDB	38.0	36.2	50.0	60.3	71.2	73.7	77.8	77.6	75.6	59.7	45.9	36.5		
	10%	WB	32.2	30.4	37.8	43.9	52.7	59.3	63.7	61.5	55.7	46.4	37.0	29.4		
		MCDB	35.3	32.9	43.8	55.7	66.7	70.9	76.1	75.7	71.6	56.3	41.2	32.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.8	19.1	22.7	24.5	22.3	24.6	24.7	24.8	21.9	18.1	16.1		
		MCDBR	18.4	20.1	27.3	33.0	31.7	28.5	31.5	32.8	35.2	30.3	26.1	20.5		
		MCWBR	14.4	15.6	16.2	16.6	13.8	12.2	12.1	11.8	14.2	15.8	16.7	15.9		
	5% WB	MCDBR	18.3	19.7	26.1	30.3	27.9	25.1	26.5	27.1	32.7	28.5	23.9	20.7		
		MCWBR	14.7	15.7	16.1	15.6	12.8	12.2	12.0	10.9	13.7	15.4	15.8	16.4		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.282	0.295	0.350	0.377	0.367	0.371	0.388	0.332	0.294	0.264	0.253		
	taud		2.367	2.376	2.444	2.298	2.265	2.356	2.370	2.285	2.428	2.509	2.477	2.383		
	Ebn at Noon		247	272	290	282	278	282	279	266	272	266	248	233		
	Edh at Noon		19	25	29	37	40	37	36	37	29	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		373	700	1135	1548	1858	1948	2102	1706	1228	747	401	283		
	RadStd		38	62	83	119	124	122	97	121	108	88	35	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page